

Case Presentation

An 89-year-old female with a history of ischemic heart failure with reduced ejection fraction (HFrEF, EF 30–35%) and atrial fibrillation (AF) presented with progressive dyspnea and orthopnea, representing her fourth hospitalization in six months for recurrent pleural effusions. Prior admissions were attributed to acute decompensated heart failure (ADHF), supported by elevated BNP and consistent symptomatic improvement following IV diuresis and thoracentesis. On this admission, chest X-ray demonstrated a large left-sided pleural effusion, similar to prior imaging. CTA chest confirmed a large left and moderate right pleural effusion with compressive atelectasis, without pulmonary embolism or mass. Initial labs revealed WBC 13.8, Hgb 12.9, Na 131, Cr 0.9, BNP 512 (previously 1200s), and hs-troponin 21 with minimal delta. During prior hospitalizations, pleural effusions were drained with transient improvement; however, pleural fluid cytology was not obtained, and the effusions were presumed to be secondary to HFrEF. Given the recurrent, unilateral predominance and discordance with expected ADHF physiology, a more comprehensive pleural fluid evaluation was pursued. Cytology subsequently revealed malignant lymphoid cells, and flow cytometry demonstrated a monoclonal B-cell population (CD19+, CD20+, CD10 dim+, lambda restricted).

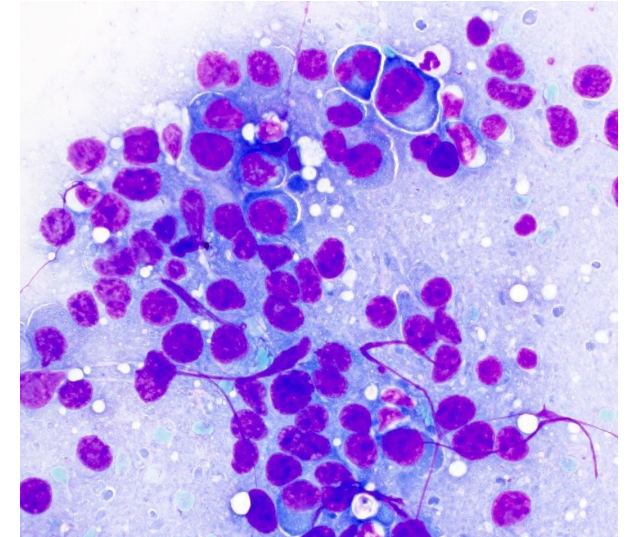
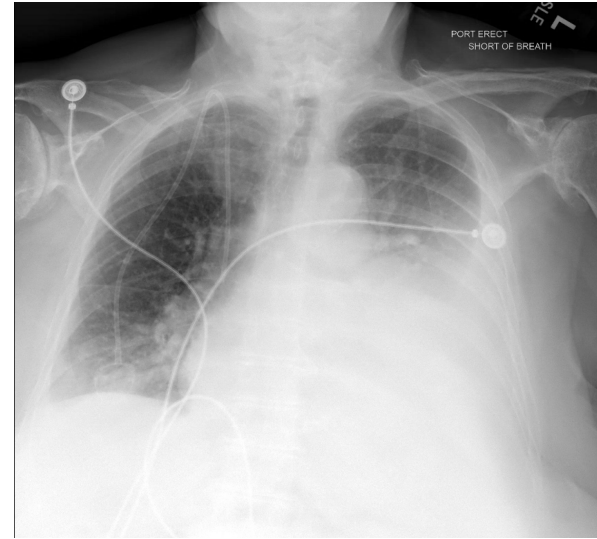


Figure 1. Chest radiograph of the patient demonstrating a large left-sided pleural effusion with associated compressive atelectasis (left) Pleural fluid cytology demonstrating large atypical lymphoid cells with prominent nucleoli and basophilic cytoplasm (right)

Background

Primary Effusion Lymphoma (PEL) is a rare and aggressive subtype of B-cell non-Hodgkin lymphoma that typically presents as malignant effusions in body cavities, most commonly the pleura, without the formation of a detectable solid tumor mass.

Symptoms: dyspnea, orthopnea, abdominal distension, pleuritic chest pain, or constitutional symptoms such as fevers and weight loss

Population Prevalence: Immunocompromised patients, especially those with Human Herpesvirus 8 (HHV-8), HIV/AIDS. Rarely found in HIV-negative, immunocompetent individuals and is commonly misattributed to Congestive Heart Failure (CHF) and Infection

Critical Treatment Protocol: median survival rate is less than 6 months. Early identification is key, especially due to resistance to standard chemotherapy regimen.

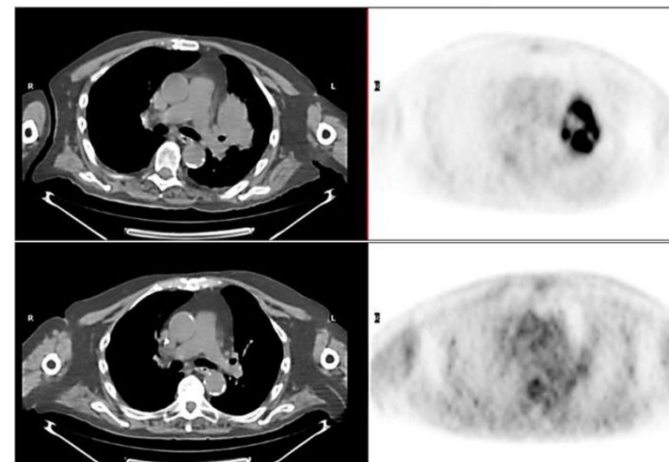


Figure 1. FDG PET-CT prior to treatment (top) and response following R-mini-CHOP therapy (bottom)

Clinical Pearls

Not all effusions are benign: Always consider cytologic evaluation.

PEL can mimic CHF: Recurrent effusions despite aggressive diuresis should raise suspicion

Cytology saves lives: Routine pleural cytology is often skipped

PEL is not exclusive to the immunocompromised: Particularly the elderly.

Early diagnosis enables intervention: In an otherwise rapidly fatal disease.